

COSMETIC INSIGHTS

Navigating Cosmetics Trends and Consumer Insights with Tableau

Data Preprocessing Steps & Business Questions with Visualization

1. Data Preprocessing Steps

- Collected raw cosmetics sales data covering 2,000 transactions and 17 fields
- Verified no missing/null values across key fields (Revenue, Units Sold, Rating, Category)
- Standardized date format (YYYY-MM-DD) for consistent time-based aggregation
- Verified categorical fields (Category, Brand, Country, Region) for consistent spelling/casing
- Connected the cleaned CSV to Tableau as a Text File data source
- Classified fields into Dimensions (categorical) and Measures (numeric) within Tableau

2. Business Questions Identified

#	Business Question	Visualization Used
1	Which product category generates the most revenue?	Bar chart — Revenue by Category
2	How does revenue trend over time?	Line chart — Revenue Trend (Monthly)
3	Which brands are top performers?	Bar chart — Top Brands by Revenue
4	Which countries contribute the most sales?	Bar chart — Revenue by Country
5	How satisfied are customers by category?	Bar chart — Average Rating by Category

3. Visualization Approach

Each business question was mapped to a single, focused chart rather than combining multiple metrics into one visual, to keep the dashboard easy to read at a glance. Bar charts were sorted in descending order to immediately surface top performers; a line chart was used specifically for the time-trend question since it best represents change over a continuous axis (date).